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TRADE RESTRICTIONS AND AGREEMENTS—TARIFFS, QUOTAS, AND EXPORT SUBSIDIES

Learning Outcome

- compare types of trade restrictions, such as tariffs, quotas, and export subsidies, and their economic implications

Trade restrictions (or trade protection) are government policies that limit the ability of domestic households and firms to trade freely with other countries. Examples of trade restrictions include tariffs, import quotas, voluntary export restraints (VER), subsidies, embargoes, and domestic content requirements. **Tariffs** are taxes that a government levies on imported goods. **Quotas** restrict the quantity of a good that can be imported into a country, generally for a specified period of time. A voluntary export restraint is similar to a quota but is imposed by the exporting country. An **export subsidy** is paid by the government to the firm when it exports a unit of a good that is being subsidized. The goal is to promote exports, but it reduces welfare by encouraging production and trade that is inconsistent with comparative advantage. **Domestic content provisions** stipulate that some percentage of the value added or components used in production should be of domestic origin. Trade restrictions are imposed by countries for several reasons, including protecting established domestic industries from foreign competition, protecting new industries from foreign competition until they mature (infant industry argument), protecting and increasing domestic employment, protecting strategic industries for national security reasons, generating revenue from tariffs (especially for developing countries), and retaliation against trade restrictions imposed by other countries.

Capital restrictions are defined as controls placed on foreigners' ability to own domestic assets or domestic residents' ability to own foreign assets. Thus, in contrast with trade restrictions, which limit the openness of goods markets, capital restrictions limit the openness of financial markets and will be addressed later.

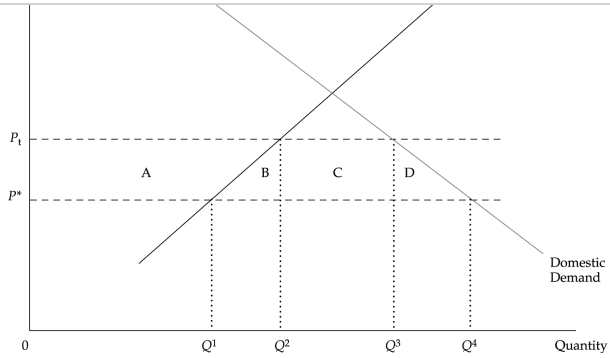
Tariffs

Tariffs are taxes that a government levies on imported goods. The primary objective of tariffs is to protect domestic industries that produce the same or similar goods. They also may aim to reduce a trade deficit. Tariffs reduce the demand for imported goods by increasing their price above the free trade price. The economic impact of a tariff on imports in a small country is illustrated in Exhibit 1. In this context, a small country is not necessarily small in size, population, or GDP. Instead, a **small country** is one that is a price taker in the world market for a product and cannot influence the world market price. For example, by many measures, Brazil is a large country, but it is a price taker in the world market for cars. A large country, however, is a large importer of the product and can exercise some influence on price in the world market. When a large country imposes a tariff, the exporter reduces the price of the good to retain some of the market share it could lose if it did not lower its price.

This reduction in price alters the terms of trade and represents a redistribution of income from the exporting country to the importing country. So, in theory, it is possible for a large country to increase its welfare by imposing a tariff if (1) its trading partner does not retaliate and (2) the deadweight loss as a result of the tariff (see below) is smaller than the benefit of improving its terms of trade. Global welfare, however, would still experience a net reduction—the large country cannot gain by imposing a tariff unless it imposes an even larger loss on its trading partner.

In Exhibit 1, the world price (free trade price) is P^*. Under free trade, domestic supply is Q^1, domestic consumption is Q^4, and imports are Q^1Q^4. After the imposition of a per-unit tariff t, the domestic price increases to P_t, which is the sum of the world price and the per-unit tariff t. At the new domestic price, domestic production increases to Q^2 and domestic consumption declines to Q^3, resulting in a reduction in imports to Q^2Q^3.

Exhibit 1: Welfare Effects of Tariff and Import Quota



▼ Expand / collapse Extended Description

The positive slope of the supply line indicates that as the price increases, producers are willing to supply more of the product, assuming other factors remain constant. The negative slope of the demand line indicates that as the price decreases, consumers are willing to buy more of the product. The point where the domestic supply and demand lines intersect is called the equilibrium point or market equilibrium.

The welfare effects of the tariff can be summarized as follows:

- Consumers suffer a loss of consumer surplus because of the increase in price. In Exhibit 1, this effect is represented by areas A + B + C + D.
- Local producers gain producer surplus from a higher price for their output. This effect is represented by area A.
- The government gains tariff revenue on imports Q^2Q^3. This effect is represented by area C.

The net welfare effect is the sum of these three effects and is summarized in Exhibit 2. The loss in consumer surplus is greater than the sum of the gain in producer surplus and government revenue and results in a deadweight loss to the country's welfare of B + D.

Exhibit 2: Welfare Effects of an Import Tariff or Quota

	Importing Country
Consumer surplus	– (A + B + C + D)
Producer surplus	+A
Tariff revenue *or* Quota rents	+C
National welfare	− B − D

Tariffs create deadweight loss because they give rise to inefficiencies on both the consumption and production side. B represents inefficiencies in production. Instead of being able to import goods at the world price P^*, tariffs encourage inefficient producers whose cost of production is greater than P^* to enter (or remain in) the market, leading to an inefficient allocation of resources. On the consumption side, tariffs prevent mutually beneficial exchanges from occurring because consumers who were willing to pay more than P^* but less than P_t are now unable to consume the good.

EXAMPLE 1

Analysis of a Tariff

South Africa manufactures 110,000 tons of paper. However, domestic demand for paper is 200,000 tons. The world price for paper is USD5.00 per ton. South Africa will import 90,000 tons of paper from the world market at free trade prices. If the South African government (a small country) decides to impose a tariff of 20 percent on paper imports, the price of imported paper will increase to USD6.00. Domestic production after the imposition of the tariff increases to 130,000 tons, while the quantity demanded declines to 170,000 tons.

1. Calculate the loss in consumer surplus arising from the imposition of the tariff on imported paper.

   Hide Solution

   Solution:

   The loss in consumer surplus = USD1 × 170,000 + 1/2 × USD1 × 30,000 = USD185,000. This calculation is represented by areas A + B + C + D in Exhibit 1.

2. Calculate the gain in producer surplus arising from the imposition of the tariff.

   Hide Solution

   Solution:

Hide Solution

Solution:

Change in government revenue = USD1 × 40,000 = USD40,000; Area C in Exhibit 1.

4. Calculate the deadweight loss arising from the imposition of the tariff.

Hide Solution

Solution:

Deadweight loss because of the tariff = 1/2 × USD1 × 20,000 + 1/2 × USD1 × 30,000 = USD 25,000; Areas B + D in Exhibit 1.

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Quotas

A quota restricts the quantity of a good that can be imported into a country, generally for a specified period of time. An **import license** specifies the quantity that can be imported. For example, the European Union operates a system of annual import quotas for steel producers who are not members of the World Trade Organization. A key difference between tariffs and quotas is that the government is able to collect the revenue generated from a tariff. This effect is uncertain under a quota. With quotas, foreign producers can often raise the price of their goods and earn greater profits than they would without the quota. These profits are called **quota rents**. In Exhibit 1, if the quota is Q^2Q^3, the equivalent tariff that will restrict imports to Q^2Q^3 is *t* and the domestic price after the quota is P_t. This is the same as the domestic price after the tariff *t* was imposed. Area C, however, is now the quota rent or profits that are likely to be captured by the foreign producer rather than tariff revenue that is captured by the domestic government. If the foreign producer or foreign government captures the quota rent, C, then the welfare loss to the importing country, represented by areas B + D + C in Exhibit 1, under a quota is greater than under the equivalent tariff. If the government of the country that imposes the quota can capture the quota rents by auctioning the import licenses for a fee, then the welfare loss under the quota is similar to that of a tariff, represented by areas B + D.

A VER is a trade barrier under which the exporting country agrees to limit its exports of the good to its trading partners to a specific number of units. The main difference between an import quota and a VER is that the former is imposed by the importer, whereas the latter is imposed by the exporter. The VER allows the quota rent resulting from the decrease in trade to be captured by the exporter (or exporting country), whereas in the case of an import quota, there is ambiguity regarding who captures the quota rents. Hence, a VER results in welfare loss in the importing country. For example, in 1981, the Japanese government imposed VERs on automobile exports to the United States.

Export Subsidies

An export subsidy is a payment by the government to a firm for each unit of a good that is exported. Its goal is to stimulate exports. But it interferes with the functioning of the free market and may distort trade away from comparative advantage. Hence, it reduces welfare. *Countervailing duties* are duties that are levied by the importing country against subsidized exports entering the country. As an example, agricultural subsidies in developed countries, notably the EU, have been a contentious issue in trade negotiations with emerging market countries and developed market countries that are agricultural exporters, such as New Zealand and Australia.

In the case of an export subsidy, the exporter has the incentive to shift sales from the domestic to the export market because it receives the international price plus the per-unit subsidy for each unit of the good exported. This scenario raises the price in the domestic market by the amount of the subsidy in the small country case (price before subsidy plus subsidy). In the large country case, the world price declines as the large country increases exports. The net welfare effect is negative in both the large and small country cases, with a larger decline in the large country case. In the large country case, the decline in world prices implies that a part of the subsidy is transferred to the foreign country, unlike in the small country case.

Exhibit 3 summarizes some of these effects.

Exhibit 3: Summary of Some of the Effects of Trade Restrictions

Panel A. Effects of Alternative Trade Policies

	Tariff	Import Quota	Export Subsidy	VER
Impact on	Importing country	Importing country	Exporting country	Importing country
Producer surplus	Increases	Increases	Increases	Increases
Consumer surplus	Decreases	Decreases	Decreases	Decreases
Government revenue	Increases	Mixed (depends on whether the quota rents are captured by the importing	Falls (government spending rises)	No change (rent to foreigners)

National welfare	Decreases in small country Could increase in large country	Decreases in small country Could increase in large country	Decreases	Decreases

Panel B. Effects of Alternative Trade Policies on Price, Production, Consumption, and Trade

	Tariff	Import Quota	Export Subsidy	VER
Impact on	Importing country	Importing country	Exporting country	Importing country
Price	Increases	Increases	Increases	Increases
Domestic consumption	Decreases	Decreases	Decreases	Decreases
Domestic production	Increases	Increases	Increases	Increases
Trade	Imports decrease	Imports decrease	Exports increase	Imports decrease

EXAMPLE 2

Analysis of Trade Restrictions

Thailand, a small country, has to decide whether to impose a tariff or a quota on the import of computers. You are considering investing in a local firm that is a major importer of computers.

1. What will be the impact of a tariff on prices, quantity produced, and quantity imported in Thailand (the importing country)?

   Hide Solution

   Solution:

   A tariff imposed by a small country, such as Thailand, raises the price of computers in the importing country, reduces the quantity imported, and increases domestic production.

2. If Thailand imposes a tariff, what will the impact be on prices in the exporting country?

   Hide Solution

   Solution:

   A tariff imposed by a small country would not change the price of computers in the exporting country.

3. How would a tariff affect consumer surplus, producer surplus, and government revenue in Thailand?

   Hide Solution

   Solution:

   When a small country imposes a tariff, it reduces consumer surplus, increases producer surplus, and increases government revenue in that country.

4. Explain whether the net welfare effect of a tariff is the same as that of a quota.

   Hide Solution

   Solution:

   The quota can lead to a greater welfare loss than a tariff if the quota rents are captured by the foreign government or foreign firms.

5. Which policy, a tariff or a quota, would be most beneficial to the local importer in which you may invest and why?

   Hide Solution

   Solution:

   A tariff will hurt importers because it will reduce their share of the computer market in Thailand. The impact of a quota depends on whether or not the importers can capture a share of the quota rents. Assuming importers can capture at least part of the rents, they will be better off with a quota.

6. If Thailand were to negotiate a VER with the countries from which it imports computers, would this be better or worse than an import quota for the local importing firm in which you may invest? Why?

   Hide Solution

   Solution:

importers.

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It is important to understand existing trade policies and the potential for policy changes that may affect return on investment. Changes in the government's trade policy can affect the pattern and value of trade and may result in changes in industry structure. These changes may have important implications for firm profitability and growth because they can affect the goods a firm can import or export; change demand for its products; affect its pricing policies; and create delays through increased paperwork, procurement of licenses, approvals, and so on. For example, changes in import policies that affect the ability of a firm to import vital inputs for production may increase the cost of production and reduce firm profitability.

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